

A Multi-Model Evaluation for Predicting Nottingham Prognostic Index from Breast Cancer Gene Expression Profiles

Group 18

Honoka Miura, Anna Cho, Celine Al-Noubani, Yuto Yagi

In this report, we describe the process of applying machine learning techniques to predict Nottingham Prognostic Index (NPI) scores from gene expression profiles for the purpose of predicting breast cancer survival using less invasive, nonsurgical methods. Specifically, we compare 2 dimensionality reduction methods (PCA, LDA) and 3 supervised learning models (logistic regression, decision tree, ResNet) for performance in binary NPI classification using accuracy, precision, F1 score, AUC-ROC, and recall metrics. Results showed relatively high but very similar score metrics across all models, indicating the presence of unexplored correlation between gene expression and NPI, and the potential of simple machine learning models being able to predict NPI scores from gene expression profiles with relatively high confidence.

1. Introduction

Breast cancer is a multifaceted disease with diverse presentations, and its heterogeneity is reflected not only in molecular signatures but also in histopathologic features. While gene expression-based classifications like PAM50 are widely used, the histologic subtype, which requires microscopy, is still heavily relied upon by clinicians. The histologic subtype is determined by pathological examination and includes classifications such as invasive ductal carcinoma, invasive lobular carcinoma, and other less common types.

Our initial objective of this project was to develop a predictive model that uses gene expression data to accurately classify breast tumors according to their histologic subtypes. Our aim was to bridge the gap between genetic expression profiles and the histological classification. Predicting the histologic subtype from gene expression can be useful since it can support diagnostics (especially when traditional microscopy assessments are ambiguous or unavailable), and offer potential biological insight.

However, we encountered severe data imbalance in our histologic subtype labels and as a result, we pivoted to predicting the Nottingham Prognostic Index (NPI), a scalar prognostic score specifically developed for breast cancer. Unlike histologic subtypes, NPI values in our dataset were more evenly distributed, allowing us to formulate a supervised learning task that is both clinically meaningful and statistically more tractable. Our revised objective, therefore, was to develop a gene expression-based model to predict NPI-defined risk groups.

NPI is calculated based on tumor size, lymph node involvement, and tumor grade, and it is widely used in clinical practice to stratify patients into risk groups with distinct survival outcomes. NPI is often calculated post-surgery after clinicians are able to get precise measurements such as tumor size. Not much research has been conducted to determine the correlation between gene expression profiles of breast cancer patients and their NPI scores in the past. By identifying the presence of this unexplored relationship between NPI and gene expression profiles and creating a machine learning model to predict NPI scores from gene expression profiles, it will provide important insight to how certain expression of genes impact breast cancer survival without/ before the need to conduct invasive surgeries, as gene expression profiles can be determined from a simple blood work.

2. Related Work

This section discusses prior work on breast cancer classification based on molecular and prognostic features, with a focus on approaches that use gene expression data to predict outcomes such as histologic subtype or prognostic indices like the Nottingham Prognostic Index (NPI).

2.1 Subtype Classification Based on Gene Expression

A foundational direction in breast cancer bioinformatics is the classification of tumors into molecular subtypes based on transcriptomic data. Perou et al. [4] introduced the concept of intrinsic molecular subtypes through gene expression profiling of breast tumors. Their findings laid the groundwork for more systematic classification systems. Building on this, Parker et al. [3] developed PAM50, a 50-gene signature classifier to assign tumors to intrinsic subtypes. PAM50 has become widely adopted in both research and clinical workflows and provides risk stratification in addition to subtype information. Further extending subtype modeling, Mathews et al. [2] proposed the Breast Cancer Consensus Subtypes (BCCS) by integrating results from multiple classification schemes across large datasets like METABRIC and TCGA. Their work highlighted immune and proliferation signals as key differentiators between subtypes. Although our project initially aimed to classify histologic subtypes from gene expression, data imbalance in rare subtypes made the task impractical, motivating a shift toward NPI prediction.

2.2 Prognostic Index Prediction Using Gene Expression

The Nottingham Prognostic Index (NPI) is a widely used clinical metric combining tumor size, grade, and lymph node status to estimate prognosis. Curtis et al. [1] conducted a large-scale integrative analysis of over 2,000 tumors from the METABRIC cohort. They uncovered ten integrative subtypes and showed strong associations between genomic/transcriptomic profiles and prognosis, laying the foundation for molecularly informed risk stratification. More recently, Zhou et al. [9] proposed a deep learning approach for directly predicting NPI categories using multi-omics data. Their model uses t-SNE to embed gene expression, CNA, and mRNA data into a 2D space and applies a ResNet classifier, achieving nearly perfect performance. Unlike their approach, which integrates multiple omics types and uses high-dimensional embeddings, we use only gene expression data and experiment with both classical models such as logistic regression and LightGBM, as well as lightweight deep learning architectures like ResNet. This allows us to explore accessible, lower-complexity alternatives suitable for omics-limited settings.

3. Methods

Our project consists of three parts: data preprocessing, predictive model development, and exploratory analysis. Beginning with data preprocessing, we first isolated the dataset csv file for columns relating to gene expression profiles and the NPI values, resulting in a total column count of 663 (489 genes + 173 mutations + 1 NPI). Despite having valuable information on mutations, we decided to eliminate the 173 gene mutations columns to focus on continuous expression levels only for simplicity. In regards to the NPI column, we have preprocessed the dataset to conduct binary classification rather than continuous regression target. This means that instead of using the provided continuous NPI values, these values were converted into zeros and ones based on the cutoff NPI of 3.4. This number derives from previous studies

that shows that 3.4 is often the NPI cutoff used to determine high vs low survival rate for patients [9]. A more detailed NPI to survival rate conversion chart is shown below: After preprocessing, we first use principal component analysis and linear discriminant analysis to determine the separability of gene

Level	Score	Survival Rate
I	from ≥ 2.0 to ≤ 2.4	93%
II	from > 2.4 to ≤ 3.4	85%
III	from > 3.4 to ≤ 5.4	70%
IV	> 5.4	50%

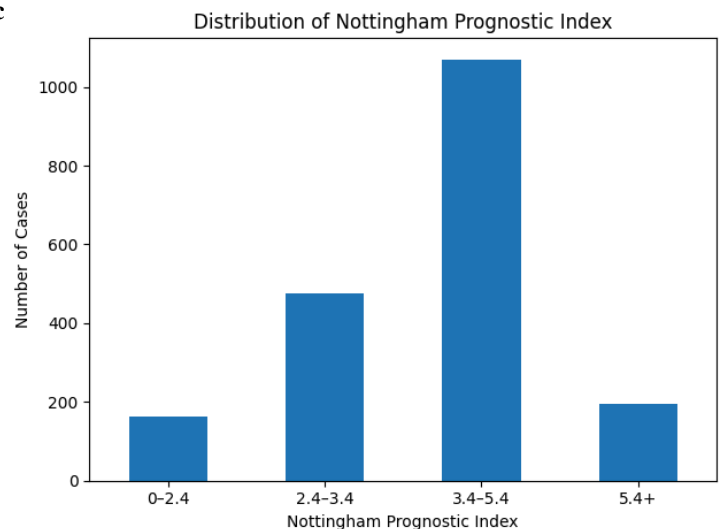
expression according to NPI. We follow these analyses by using three machine learning models: logistic regression, decision tree, and residual networks to classify patients into NPI categories based on gene expression. Detailed explanations of each model are mentioned below under Experiments.

Table 1. Nottingham Prognostic Index Scores and their Survival Rate Conversions

4. Data

Figure 1. Nottingham Prognostic Index Distribution Across Patients in Dataset

The data used in this project comes from the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) database collected by Professor Carlos Caldas from Cambridge Research Institute and Professor Sam Aparicio from the British Columbia Cancer Centre in Canada. This dataset contains sequencing data from 1980 breast cancer patients as well as other demographics and attributes such as age at prognosis, cancer type, mutation count, and tumor size. The dataset contains mRNA gene expression across 489 cancer-related genes and mutation data for 173 genes. The attributes of interest in this dataset for the purposes of this project is the Nottingham prognostic index expressed as a decimal index value.



4.4 Histological Type

The histologic subtype distribution was significantly skewed. Out of 1980 samples, 1,500 (79.4%) are invasive ductal carcinoma, 207 (11.0%) are mixed ductal-lobular, and 142 (7.5%) are invasive lobular carcinoma. The remaining subtypes had much fewer samples with mixed mucinous carcinoma having 22

cases (1.2%) and metaplastic breast cancer had just 1 case, ($<0.1\%$). Additionally around 17 cases (0.9%) had missing values. This strongly skewed data, particularly the heavily limited sample size for mixed mucinous carcinoma and metaplastic breast cancer makes it difficult for any model to learn the underlying pattern from this dataset. Because the model sees overwhelmingly more invasive ductal carcinoma examples and not enough of the rare cases, it becomes heavily biased toward predicting that class. In our initial experimentation with decision-tree based models, the model could not reliably learn their distinguishing patterns. Due to this extreme class imbalance along with limited size of samples, we have decided to pivot away from predicting histological types in our project.

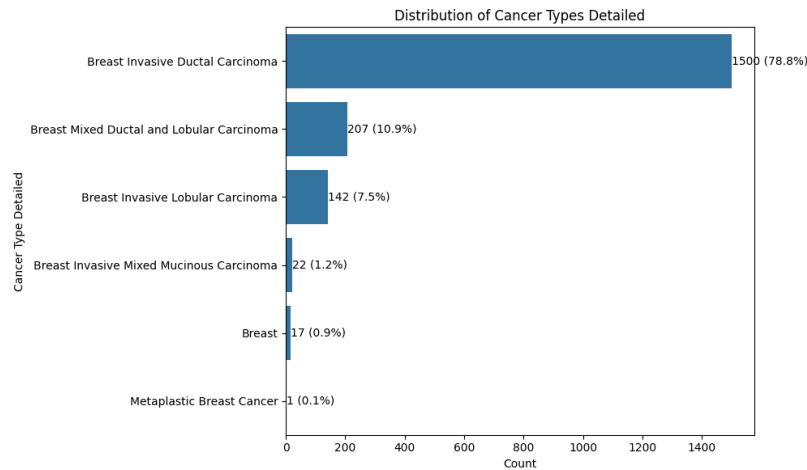


Figure 2. Distribution of Histologic Subtypes Across Patients in Dataset

5. Experiments

5.1 Dimensionality Reduction

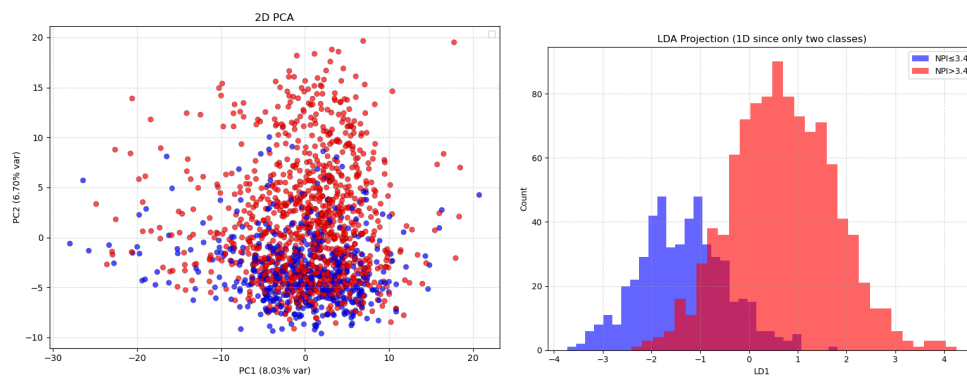


Figure 3. PCA (left) and LDA (right) Projections After Dimensionality Reduction

5.1.1 Principal component analysis (PCA)

Since our dataset contains high dimensionality (600+ features), we applied dimensionality reduction technique as a part of our exploratory data analysis to assess if it can effectively reduce the dimension. First, we tried PCA, which simplifies to find component axes that maximize the variance. Our results are shown in the figure below. The figure shows a general pattern of NPI above 3.4 clustering near higher values of PC2 and NPI below 3.4 clustering around lower values of PC2. However, there is a significant overlap between the two clusters, with almost all samples for NPI below 3.4 located very close to samples from the other class. This suggests that the components found through PCA are not very effective in capturing patterns that distinguish the two classes.

5.1.2 Linear Discriminant Analysis (LDA)

In order to align more closely with our original objective of the binary classification, we also applied LDA to our dataset. Unlike PCA, LDA selects component axes to maximize class separation. For this reason, LDA is more promising that they can find a better component that separates the two classes.

Figure below shows the distribution of the single Linear Discriminant obtained when projecting our standardized 600+ features onto the one axis that most strongly separates the two NPI groups. The histogram for the low-NPI group (≤ 3.4 , in blue) is almost entirely to the left of the origin (centered around -1.5), whereas the high-NPI group (> 3.4 , in red) lies to the right (centered near $+1.0$), with a small overlap around LD between -2 to $+1$.

Because LDA for two classes produces exactly one discriminant, this separation indicates that the RNA gene expression features contain a strong pattern that can be used to separate two classes of NPI. This also suggests that there may exist a clear linear decision boundary in our dataset.

Additionally, in order to further understand how the LDA separated the two classes, we extracted the feature importance from the model. The figure below shows the top 15 features with highest importance. It is notable that the top three genes, including *gata3*, *mmp11*, *col12a1*, have particularly strong importance. Quick literature review reveals that these genes are indeed associated with prognosis of breast cancer. Past researches found that *gata3* is associated with favorable pathologic features of breast cancer (Voduc, Cheang, & Nielsen, 2008), *mmp11* promote the migration of HER2+ breast cancer cells (Kang, et.al., 2022), and *col12a1* has shown to have a potential to be a prognostic biomarker (Yan, et.al., 2023). These suggest that the important features extracted from LDA based on NPI classification have correlation with prognosis of breast cancer. More research along this line may help identify unknown prognostic biomarkers.

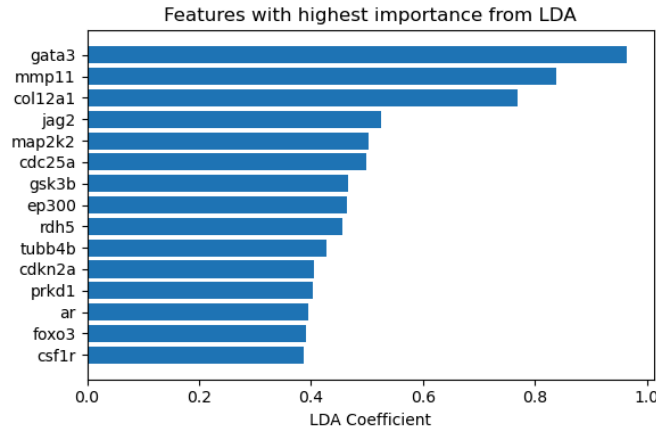


Figure 4. Bar Graph Representing Top 15 Gene Features With Highest Classification Importance

5.2 Logistic Regression

Model Setup

This study implemented three logistic regression models to predict the NPI binary classification (Low Risk ≤ 3.4 vs High Risk > 3.4) using 488 numeric gene expression features from the METABRIC dataset. The baseline model used default parameters ($C=1.0$, L2 regularization) and achieved moderate performance. The fine-tuned model employed grid search with 5-fold cross-validation across 24 combinations of hyperparameters, testing regularization strengths (C values from 0.001 to 100), regularization types (L1 and L2), and class weighting options. This optimization process identified optimal hyperparameters ($C=0.1$, L1 regularization, no class weighting) that substantially improved performance. The third model used L1 regularization-based feature selection, reducing the feature space from 488 to just 8 genes while maintaining comparable predictive power. All models used a consistent 80/20 train/test split with stratification to handle the class imbalance (66.4% high-risk, 33.6% low-risk), and features were standardized using StandardScaler prior to model training.

Results

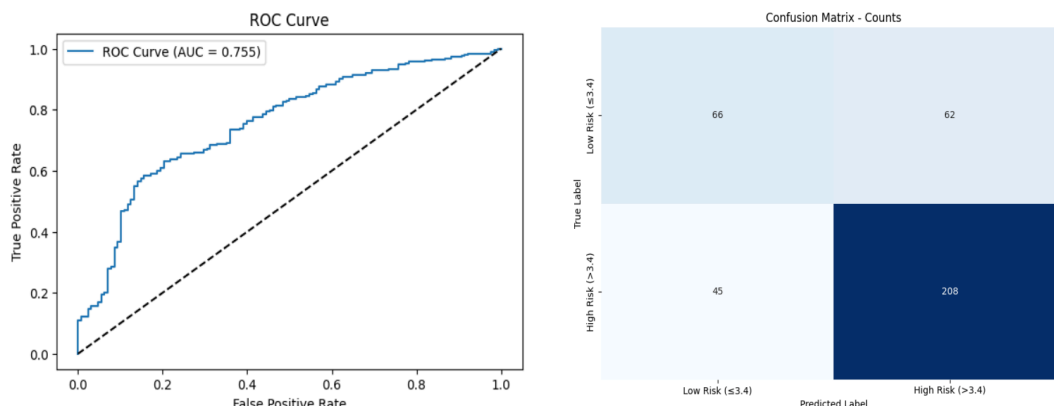


Figure 5. Evaluation Using ROC Curve (left) and Confusion Matrix (right) For Logistic Regression

The fine-tuned logistic regression model demonstrated significant improvement over the baseline, achieving 71.9% accuracy with hyperparameters of $C=0.1$, L1 regularization, and no class weighting. The model exhibited strong recall (82.2%), indicating excellent sensitivity in identifying high-risk patients, while maintaining good precision (77.0%). The confusion matrix reveals that the model correctly identified 208 high-risk patients (82.2% of actual high-risk cases) while misclassifying only 45 high-risk patients as low-risk. However, the model showed higher false positive rates, misclassifying 48.4% of low-risk patients as high-risk. This performance profile makes the model particularly suitable for screening purposes, where missing high-risk patients would be more concerning than overtreating low-risk ones. The ROC-AUC of 0.755 indicates good discriminative ability, and the model's optimal decision threshold of 0.564 provides the best balance between precision and recall. Notably, the model achieves these results while maintaining good generalization, with cross-validation showing stable performance of $73.5\% \pm 3.2\%$ accuracy across different data subsets.

5.3 Decision Trees (LightGBM)

Model Setup

Given the high dimensionality of our data set, we decided to apply a decision tree based model to our task since they are known to handle high dimensionality very well. We used lightGBM library for our decision tree based model due to its efficiency and wide usage.

We used all gene expression data and mutation data as features, and tuned hyperparameters across a large search space to ensure we reach the maximum accuracy for this model architecture.

Results

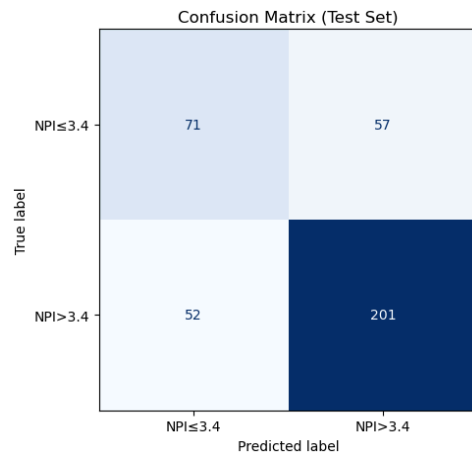


Figure 6. Evaluation Using Confusion Matrix For Decision Tree

Overall, the model reached an accuracy of 71.4%. For the first class of NPI below 3.4, it had a f1-score of 57%, with 58% precision and 55% recall. For the second class of NPI above 3.4, it had a f1-score of 78% with 79% precision and 79% recall. This shows that the model has an equal balance between precision and recall across both classes, but consistently more accurately for the class of NPI above 3.4. This suggests that even with sample weight adjustment to counter the data imbalance, the decision tree based model may still have difficulty learning the pattern for underrepresented classes.

This result seems unremarkable compared to past literatures which gained accuracy above 90% using larger dataset and more complex models. We think that this result is reasonable given that we had a limited data size of less than 2000 samples, and also used a quite simple architecture of Lightgbm.

5.4 Convolutional Neural Networks (ResNet)

Model Setup

A deep learning architecture using ResNet was implemented for the last model to classify breast cancer gene expression profiles into NPI categories. Deep learning networks such as ResNet are able to capture hundreds of features and are suitable for situations like this where the network must learn subtle nonlinear interactions between many genes. Similar to the decision tree, a weighted random sampler was used to balance class representation between the two NPI classifications. The ResNet also relies on skip connections to reduce vanishing gradient problems and overfitting. Finally, hyperparameters were fine tuned to produce the final metrics with the best parameters being hidden dimension of 512, learning rate of 0.005, batch size of 32, and epoch size of 50.

Results

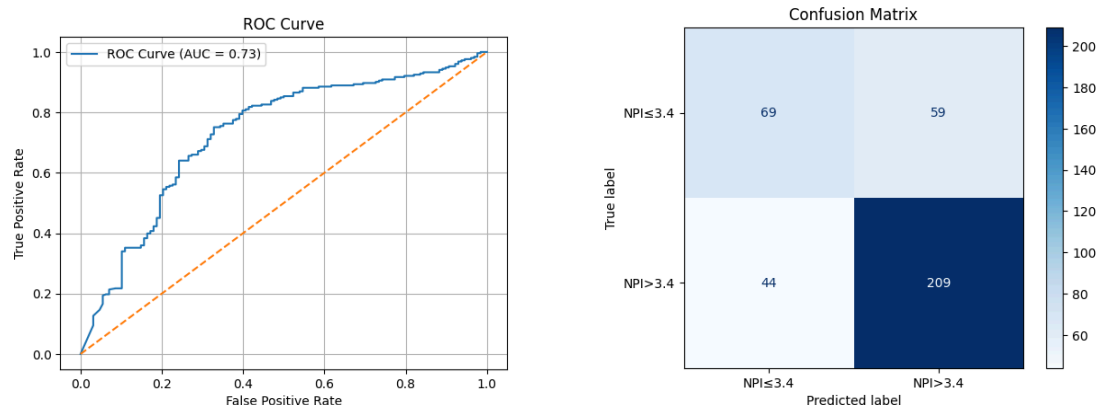


Figure 7. Evaluation Using ROC Curve (left) and Confusion Matrix (right) For ResNet

Overall, the maximum accuracy obtained was 0.7297, or 72.97%. The NPI below 3.4 class showed a precision score of 0.61, recall of 0.54, and F1 of 0.57. On the other hand, the NPI above 3.4 class showed a higher score for all metrics compared to the other class with precision of 0.78, recall of 0.83, and F1 of 0.80. Similar to decision trees, despite using a weighted random sampler to account for data imbalance, the model still performs less effectively for underrepresented classes. The resulting model metrics have decent values and show a sign of slight potential for improvement. However, seeing how similar the model performed compared to the decision tree, the ResNet model may have reached maximum capacity and the low performance may be due to limitations from the dataset. Decision Tree and ResNet have a foundationally different approach to classification, and therefore, we expected to see ResNet perform much better than the rest of the models, as ResNet handles nonlinear, complex, high feature data better than others. Despite this, we see similar resulting metrics.

6. Evaluation and Discussion

Model	Accuracy	Recall	Precision	AUC	F1
Logistic Regression	0.719	0.822	0.770	0.755	0.795
Decision Tree (lightgbm)	0.714	0.78	0.79	0.74	0.79
ResNet	0.7297	0.8261	0.7799	0.7289	0.8023

Table 2. Final Metric Comparison Between Supervised Learning Models

The performance of our predictive models were rigorously evaluated using a suite of standard classification metrics to ensure both accuracy and reliability. We assess overall prediction performance through accuracy, precision, recall, F1-score, and ROC-AUC. Despite differences in model complexity, all three classifiers achieved comparable performance, with accuracy ranging from 71.4% to 72.9% and

F1 scores from 0.79 to 0.80. This narrow margin suggests that the gene expression patterns separating low- and high-risk NPI groups are partially linearly separable, allowing even simpler models to perform effectively.

Logistic regression achieved an accuracy of 71.9%, recall of 82.2%, and an AUC of 0.755, offering the best balance between performance and interpretability. It also highlighted key predictive genes, making it suitable for clinical use where model transparency is important. LightGBM yielded similar results (accuracy 71.4%, recall 78%, AUC 0.74), showing strong performance despite the high-dimensional input. However, it struggled slightly more with the low-risk class, likely due to class imbalance. ResNet, the most complex model, achieved the highest accuracy (72.97%) and recall (82.6%), along with an F1 score of 0.8023. However, its performance advantage over simpler models was marginal, and its limited interpretability makes it less ideal for clinical deployment.

Interestingly, all models performed consistently better on the high-risk group ($NPI > 3.4$), suggesting more distinct gene expression patterns or greater label consistency in that class. LDA projections further supported this, revealing a clear linear separation between risk groups and highlighting biologically meaningful genes like GATA3, MMP11, and COL12A1.

Overall, the results emphasize that model complexity does not guarantee better performance, especially in omics-limited settings. Instead, interpretable models like logistic regression can deliver strong predictive value while also offering insight into underlying biological mechanisms.

7. Limitations of the current work and future direction

Despite these encouraging results, our study has several limitations. First, the modest sample size (<2,000 patients) and reliance on a single cohort may limit generalizability, and hence, the external validation on independent datasets is needed. Second, we trained all models solely on mRNA expression, omitting complementary data types such as mutation calls and copy-number alterations that could enrich predictive power. Third, although the ResNet architecture improved specificity, its deep, non-linear architecture and residual connections make it challenging to interpret its decisions by simple weight inspection. To overcome these challenges, future studies should incorporate multi-omics integration, apply model-agnostic interpretability techniques to quantify each gene's contribution to individual predictions, and explore ensemble strategies that combine complementary strengths of logistic regression, LightGBM, and ResNet.

8. Contributions

Celine=Logistic Regression, Yuto= Dimensionality Reduction & Decision Tree, Honoka & Anna= ResNet, All=Data Preparation & Report

9. Source Code

All of our source code can be found at the following link:
<https://github.com/gatech/yyagi6/cse7580-project.git>

Reference

1. Curtis, C., Shah, S. P., Chin, S. F., Turashvili, G., Rueda, O. M., Dunning, M. J., ... & Aparicio, S. A. (2012). The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups. *Nature*, 486(7403), 346–352. <https://doi.org/10.1038/s41523-021-00345-2>
2. Mathews, J. C., Nadeem, S., Levine, A. J., Pouryahya, M., Deasy, J. O., & Tannenbaum, A. (2021). Breast cancer consensus subtypes: A system for subtyping breast cancer tumors based on gene expression. *NPJ Breast Cancer*, 7(1), 136. <https://doi.org/10.1038/s41523-021-00345-2>
3. Parker, J. S., Mullins, M., Cheang, M. C., Leung, S., Voduc, D., Vickery, T., ... & Perou, C. M. (2009). Supervised risk predictor of breast cancer based on intrinsic subtypes. *Journal of Clinical Oncology*, 27(8), 1160–1167. <https://doi.org/10.1200/JCO.2008.18.1370>
4. Perou, C. M., Sorlie, T., Eisen, M. B., van de Rijn, M., Jeffrey, S. S., Rees, C. A., ... & Botstein, D. (2000). Molecular portraits of human breast tumours. *Nature*, 406(6797), 747–752. <https://doi.org/10.1038/35021093>
5. Pereira, B., Chin, SF., Rueda, O. *et al.* The somatic mutation profiles of 2,433 breast cancers refine their genomic and transcriptomic landscapes. *Nat Commun* 7, 11479 (2016). <https://doi.org/10.1038/ncomms11479>
6. Voduc D, Cheang M, Nielsen T. GATA-3 expression in breast cancer has a strong association with estrogen receptor but lacks independent prognostic value. *Cancer Epidemiol Biomarkers Prev*. 2008 Feb;17(2):365-73. doi: 10.1158/1055-9965.EPI-06-1090. PMID: 18268121
7. Kang SU, Cho SY, Jeong H, Han J, Chae HY, Yang H, Sung CO, Choi YL, Shin YK, Kwon MJ. Matrix metalloproteinase 11 (MMP11) in macrophages promotes the migration of HER2-positive breast cancer cells and monocyte recruitment through CCL2-CCR2 signaling. *Lab Invest*. 2022 Apr;102(4):376-390. doi: 10.1038/s41374-021-00699-y. Epub 2021 Nov 13. PMID: 34775491
8. Yan Y, Liang Q, Liu Y, Zhou S, Xu Z. COL12A1 as a prognostic biomarker links immunotherapy response in breast cancer. *Endocr Relat Cancer*. 2023 Apr 3;30(5):e230012. doi: 10.1530/ERC-23-0012. PMID: 36877531.
9. Zhou, L., Rueda, M., & Alkhateeb, A. (2022). Classification of Breast Cancer Nottingham Prognostic Index Using High-Dimensional Embedding and Residual Neural Network. *Cancers*, 14(4), 934. <https://doi.org/10.3390/cancers14040934>